$$
\begin{aligned}
v'^2 &= \frac{v^2 - 2uv + u^2}{1 - 2uv + u^2 v^2}, \\
1 - v'^2 &= \frac{1 - 2uv + u^2 v^2 - v^2 + 2uv - u^2}{1 - 2uv + u^2 v^2} \\
&= \frac{1 - v^2 - u^2 + u^2 v^2}{1 - 2uv + u^2 v^2} \\
&= \frac{(1 - v^2)(1 - u^2)}{(1 - uv)^2}.
\end{aligned}
$$

Therefore

$$
\frac{1}{\sqrt{1 - v'^2}} = \frac{1 - uv}{\sqrt{1 - v^2}\sqrt{1 - u^2}}. \tag{5.9}
$$

The energy E' is then simply m_0 times the above expression. But we want to express the energy in terms of the unprimed energy and momentum, and we note that

$$
E' = \frac{m_0 - m_0 uv}{\sqrt{1 - v^2}\sqrt{1 - u^2}} = \frac{(m_0/\sqrt{1 - v^2}) - (m_0 v/\sqrt{1 - v^2})\,u}{\sqrt{1 - u^2}},
$$

or

$$
E' = \frac{E - up_x}{\sqrt{1 - u^2}}, \tag{5.10}
$$

which we recognize as being exactly of the same form as

$$
t' = \frac{t - ux}{\sqrt{1 - u^2}}.
$$

Next we must find the new momentum p'_x. This is just the energy E' times v', and is also simply expressed in terms of E and p:

$$
p'_x = E'v' = \frac{m_0(1 - uv)}{\sqrt{1 - v^2}\sqrt{1 - u^2}} \cdot \frac{v - u}{(1 - uv)} = \frac{m_0 v - m_0 u}{\sqrt{1 - v^2}\sqrt{1 - u^2}}
$$

Thus

$$
p'_x = \frac{p_x - uE}{\sqrt{1 - u^2}}, \tag{5.11}
$$